

Minimally Invasive Surgery for Sparse Neural Networks in Contrastive Manner

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Abstract

With the development of deep learning, neural networks tend to be deeper and larger to achieve good performance. Trained models are more compute-intensive and memory-intensive, which lead to the big challenges on memory bandwidth, storage, latency, and throughput. In this paper, we propose the neural network compression method named minimally invasive surgery. Different from traditional model compression and knowledge distillation methods, the proposed method refers to the minimally invasive surgery principle. It learns the principal features from a pair of dense and compressed models in a contrastive manner. It also optimizes the neural networks to meet the specific hardware acceleration requirements. Through qualitative, quantitative, and ablation experiments, the proposed method shows a compelling performance, acceleration, and generalization in various tasks.

1. Introduction

Deep learning technologies promote performance in various applications like computer vision, natural language processing, autonomous driving, recommendation system, etc. The promising performance is achieved by deeper and larger neural networks. For example, the classical architectures in convolutional neural networks like VGG-19 [41], ResNeXt-101 [49], SENet-154 [17] has 143.67, 83.46, and 115.09 million parameters, respectively. Google's neural machine translation model [48] has about 210 million parameters. The popular language understanding model BERT [8] has about 340 million parameters. The deep learning recommendation model (DLRM) [29] has about 540 million parameters.

Neural networks with a huge amount of parameters have some shortcomings [56]. First of all, the large neural network is very compute-intensive. In the network evaluation process, inference costs a lot of time even the network is running on dedicated acceleration hardware like GPU [31] [32] or TPU [40]. We can enlarge the batch size to help improve the throughput of large neural networks. But

the latency is still a problem. In fact, whenever we interact with phones or computers, we are very sensitive to the latency of the interaction. We don't like to wait for an application to launch or for the web-page to load search results. Moreover, we are especially sensitive in realtime interactions such as speech recognition and autonomous driving systems. Secondly, the large neural network is memory-intensive on mobile devices as well as in the server environment. Storage and loading the large neural network to compute inference results consume a large amount of energy. Due to the limitations on application sizes, download time and launch speed, transfer and storage of large models is especially a challenge in the mobile environment.

Compressing the large neural network to a smaller version can bring benefits to more efficient computation, memory, and energy consumptions. But at the meanwhile, how to keep the accuracy of the original neural network during compression needs to be investigated. A common method of neural model compression is network pruning [12]: setting the weights with small magnitude values of a pre-trained network to zero and fine-tuning the remaining weights to try to recover accuracy. For the aggressive network pruning tasks, knowledge distillation [15] is often used as the auxiliary method to improve the accuracy of the pruned network. A complementary method of neural model compression is quantization. Changing fundamental data types adds the ability to accelerate the arithmetic operations, both in training [28] and inference processes [20].

In this work, we explore a neural network compression method based on the knowledge extracted from a pair of dense and compressed models. We named this method as **Minimally Invasive Surgery(MIS)** because it is inspired by the principle and process of the real minimally invasive surgery. We apply the **MIS** technique to several networks and tasks to show generality in supervised and unsupervised learning. Our main contributions include:

- We prove that **MIS** has better performance than knowledge distillation and network compression methods.
- We provide the theoretical demonstration of **MIS** from information entropy and Bayes perspectives.
- We show that **MIS** technique can apply to various net-